

Tracking Performance Drift in Large Language Models Across Successive Version Updates: A Survey

VEDARTHA SAI Y, Department of Information Technology, IIITA, India

Large Language Models (LLMs) are increasingly integrated into critical enterprise, academic, and software systems [1]. Unlike conventional software, these models are often continuously updated; model weights, post-training procedures, safety mechanisms, and serving infrastructure may change without complete public documentation [2]. Such changes can produce **performance drift**, defined as a gradual or sudden change in a model's observable behavior when the same or equivalent inputs are presented across versions or points in time [2]. This paper examines performance drift in LLMs across mathematical reasoning, programming, instruction following, truthfulness, safety, alignment, and multi-turn interaction [2, 4]. Previous longitudinal research has demonstrated measurable behavioral changes in systems such as GPT-3.5 and GPT-4 across successive updates [2]. This paper synthesizes that work with established evaluation frameworks, including MMLU, HumanEval, TruthfulQA, HELM, and MT-Bench [3, 5–7, 13]. The analysis demonstrates that LLM performance is multidimensional rather than a single static capability score: an update may improve one dimension while producing changes or regressions in another [6]. The paper further evaluates behavioral fingerprinting, longitudinal benchmarking, executable evaluation, human assessment, and LLM-based judging as approaches to monitoring drift [13]. Major challenges include the black-box nature of proprietary models, stochastic generation, benchmark contamination, the difficulty of establishing ground truth for open-ended tasks, and the complexity of multi-turn interactions [4]. The paper concludes that continuous, multidimensional, and reproducible evaluation is essential for maintaining reliability when LLMs are deployed as evolving services [2].

CCS Concepts: • **Computing methodologies** → **Natural language processing**: *Knowledge representation and reasoning*; • **Software and its engineering** → *Software maintenance tools*.

Additional Key Words and Phrases: Large Language Models, Performance Drift, Behavioral Drift, Model Updates, Longitudinal Evaluation, Benchmarking, AI Reliability, Instruction Following, AI Alignment

ACM Reference Format:

Vedarthai Sai Y. 2026. Tracking Performance Drift in Large Language Models Across Successive Version Updates: A Survey. 1, 1 (September 2026), 11 pages. <https://doi.org/10.1145/3561048>

Author's Contact Information: Vedarthai Sai Y, mhm2026013@iiita.ac.in, Department of Information Technology, IIITA, Prayagraj, India.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM XXXX-XXXX/2026/9-ART
<https://doi.org/10.1145/3561048>

1 INTRODUCTION

The widespread adoption of Large Language Models (LLMs) has fundamentally transformed natural language processing, automated reasoning, information retrieval, software development, and decision-support systems [1]. Models such as GPT-3.5 and GPT-4 have become components of commercial applications ranging from coding assistants to conversational interfaces and automated information-processing systems [8]. However, unlike traditional software packages, many proprietary LLMs are delivered through application programming interfaces (APIs), meaning that users may interact with continuously updated systems without having direct access to the underlying model weights, training data, or post-training procedures [2].

The continuous updating of these systems introduces an important reliability concern: **performance drift** [2]. In this paper, performance drift refers to changes in the observable behavior of an LLM across different versions or time periods when the evaluation task and conditions are intended to remain constant [2]. Such changes may represent improvements, regressions, or trade-offs between different capabilities [6]. A model update intended to improve safety, factuality, or instruction following may simultaneously alter coding behavior, formatting, reasoning performance, or conversational style [10].

The problem is particularly important because conventional software regression testing assumes that a fixed program should produce stable results for identical inputs [2]. LLMs do not necessarily exhibit such deterministic behavior [1]. Their outputs can vary because of sampling, model configuration, prompt formulation, and changes to the model or serving system [2]. When proprietary APIs alter their internal mechanics without public release notes, downstream enterprise applications experience unexpected breakages, corrupted outputs, and system latency shifts. Consequently, monitoring an LLM requires statistical and behavioral evaluation rather than simply checking whether a particular output exactly matches a previous output [6].

A significant longitudinal investigation was conducted by Chen et al. [2], who examined changes in ChatGPT behavior between March and June 2023. Their findings demonstrated substantial changes in mathematical problem solving, coding, and responses to sensitive questions [2]. In particular, the study reported that GPT-4’s performance on a prime-number identification task changed considerably between the two evaluation periods [2]. Such findings demonstrate that model performance cannot necessarily be assumed to remain constant merely because the model retains the same public designation [2].

The problem extends beyond individual benchmarks [6]. Established evaluation frameworks have demonstrated that LLM performance is inherently multidimensional [6]. MMLU evaluates knowledge and reasoning across numerous academic subjects [5], HumanEval evaluates functional correctness in code generation [3], and TruthfulQA examines whether language models reproduce common misconceptions [7]. HELM further emphasizes the importance of evaluating models across multiple scenarios and metrics rather than relying on a single aggregate performance measure [6].

Therefore, the objective of this paper is to examine how LLM performance can drift across successive updates and to identify methodological approaches for detecting, measuring, and mitigating such changes [2]. The remainder of this paper is structured as follows: Section 2 surveys related literature and foundational evaluation frameworks; Section 3 presents thematic discussions alongside mathematical formulations of performance drift; Section 4 compares mitigation strategies and continuous evaluation architectures; Section 5 details operational limitations; Section 6 outlines research gaps; and Section 7 provides concluding remarks. Figure 1 provides a taxonomy mapping the multi-faceted dimensions of LLM performance drift examined throughout this work.

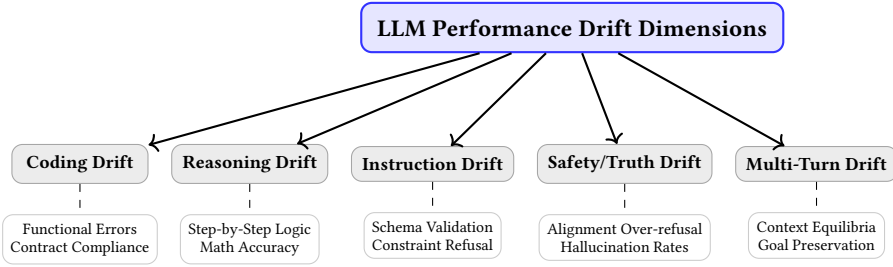


Fig. 1. Taxonomy of LLM Performance Drift across key operational capability dimensions.

2 BACKGROUND AND RELATED WORK

2.1 LLM Evaluation and Benchmarking

The evaluation of LLMs has traditionally focused on measuring performance at a particular point in time [6]. Benchmarks provide standardized tasks that allow researchers to compare models under controlled conditions [5]. However, longitudinal drift evaluation introduces a different requirement: the same model or service must be measured repeatedly over time [2].

MMLU, introduced by Hendrycks et al. [5], evaluates models across 57 subjects spanning areas such as mathematics, computer science, law, history, and other academic disciplines. Its broad coverage makes it useful for determining whether an observed performance change is restricted to a specific domain or represents a broader capability shift.

Similarly, HumanEval was introduced by Chen et al. [3] to evaluate code-generation systems through functional correctness. Rather than judging code solely according to textual similarity, HumanEval uses executable tests to determine whether generated programs solve the intended programming problems [3]. This distinction is valuable for drift analysis because it separates changes in code presentation from changes in actual programming capability.

HELM, proposed by Liang et al. [6], provides a broader evaluation framework based on multiple scenarios and metrics. The framework considers dimensions including accuracy, calibration, robustness, fairness, bias, toxicity, and efficiency [6]. This multidimensional perspective is particularly relevant to performance drift because a model update can simultaneously improve and worsen different properties.

2.2 Behavioral Changes in Proprietary Models

Chen et al. [2] provide one of the most directly relevant longitudinal investigations. Their research examined ChatGPT at multiple points in time and found that the behavior of GPT-3.5 and GPT-4 changed substantially across selected tasks [2]. The study reported differences in mathematical reasoning, code generation, and responses to sensitive questions.

The significance of this work lies in demonstrating that LLM evaluation results are inherently time-dependent [2]. A benchmark result obtained in March cannot necessarily be interpreted as a permanent characteristic of the service several months later [2]. This creates a strict requirement for timestamped evaluation and version-aware reporting.

Liu et al. [8] similarly examine the capabilities of ChatGPT across a range of tasks and emphasize that apparent model capabilities depend heavily on the particular evaluation setting. Together, these findings support the need to distinguish between a model’s nominal identity and its observed behavior at a particular point in time [2].

2.3 Instruction Following, Human Feedback, and the Alignment Tax

Post-training procedures can significantly alter model behavior [10]. Ouyang et al. [10] demonstrated that training language models using human feedback can substantially improve their ability to follow instructions and align their responses with human preferences. This research is important for understanding drift because updates to instruction-following and alignment mechanisms can produce unintended changes beyond the specific behavior targeted by an update.

The GPT-4 technical report similarly describes extensive post-training and alignment procedures used in the development of GPT-4 [9]. Because these processes alter the model’s behavior, changes in alignment should be considered an important component of longitudinal performance evaluation [9]. Often, enhancing safety constraints introduces an “alignment tax,” where models become overly conservative, thereby degrading performance on benign tasks or complex reasoning structures.

2.4 Comparative Summary of Existing Literature

Table 1 provides a systematic comparative summary of key research approaches, evaluation frameworks, primary datasets, core findings, and intrinsic limitations in existing LLM evaluation literature.

Table 1. Comparative summary of key research frameworks and evaluation literature on LLM performance and drift.

Study / Framework	Core Methodology	Datasets / Tasks	Key Findings	Primary Limitations
Chen et al. [2]	Longitudinal API tracking (Mar vs Jun 2023)	Prime numbers, HumanEval, sensitive prompts	Significant drift across updates (e.g., GPT-4 math drop from 84% to 51%)	Black-box API access; lacks weight-level causal tracing.
MMLU [5]	Multiple-choice standard evaluation	57 academic subjects (STEM, Humanities, Law)	Effective broad capability baseline across domain knowledge	Susceptible to dataset contamination; single-turn static.
HumanEval [3]	Executable unit-testing verification	164 hand-written Python coding problems	Separates code style/formatting from functional correctness	Limited task complexity; single language scope.
HELM [6]	Holistic multi-metric evaluation	42 scenarios across accuracy, bias, toxicity	LLM performance is multi-faceted; optimizations induce trade-offs	High computational cost to execute continuously.
TruthfulQA [7]	Misconception prompt probing	817 questions triggering human false beliefs	Larger models can mimic popular misconceptions more strongly	Static questions can be overfitted during post-training.
MT-Bench [13]	LLM-as-a-judge multi-turn scoring	80 multi-turn conversational scenarios	Automated LLMs can effectively judge open-ended turn quality	Position, verbosity, and self-enhancement judge biases.
Dongre et al. [4]	Context equilibrium dynamics	Extended conversational context streams	Long context interactions suffer non-linear context degradation	Evaluated on restricted conversational patterns.

3 MAIN THEMATIC AND TECHNICAL DISCUSSION

3.1 Degradation in Step-by-Step Reasoning

Reasoning is one of the most important and difficult dimensions of LLM performance to monitor [2]. Chen et al. [2] demonstrated that GPT-4’s accuracy on a prime-number classification task changed substantially between March and June 2023. The study provides a clear example of how reasoning performance can vary even over relatively short periods [2].

Such observations challenge the assumption that newer versions necessarily represent uniform improvements [2]. An update can subtly modify how a model interprets instructions or produces

intermediate reasoning (e.g., Chain-of-Thought), potentially destroying performance on specific logical tasks.

Broad benchmarks such as MMLU [5] can help determine whether changes are localized or widespread. However, aggregate benchmark scores may conceal individual task regressions [6]. A model could maintain or improve its overall score while experiencing a substantial decline on a small but operationally important subset of tasks [2].

Consequently, longitudinal reasoning evaluation should record both aggregate performance and task-level performance [6]. Confidence intervals and repeated trials should also be considered because stochastic generation can produce different answers under nominally identical conditions [1]. Figure 2 illustrates this longitudinal discrepancy. Mathematical formalization of this behavior is elaborated in Section 3.2.

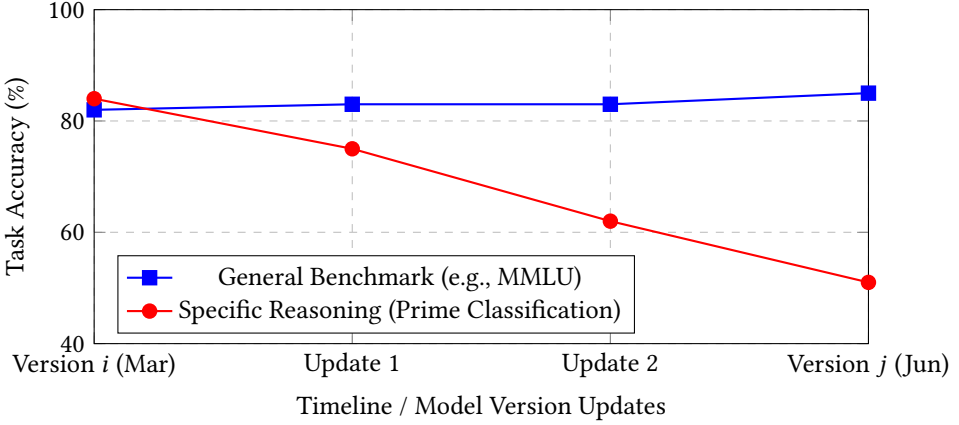


Fig. 2. An illustration of longitudinal performance drift. As noted by Chen et al. [2], aggregate benchmark scores may remain stable or improve across version updates, while specific capabilities (such as step-by-step reasoning on prime-number classification) can degrade sharply from 84% to 51%.

3.2 Mathematical Formulation of Performance Drift

To quantify changes in LLM behavior across successive versions, performance drift can be represented using mathematical measures [2]. Let P_i denote the performance of a model on a particular evaluation task at an earlier version or time point, and let P_j denote its performance at a later version or time point. The relative performance drift between the two versions can be defined as shown in (1):

$$D_{i,j} = \frac{P_j - P_i}{P_i} \times 100 \quad (1)$$

where $D_{i,j}$ represents the percentage change in performance between versions i and j . A negative value indicates a reduction in performance, whereas a positive value indicates an improvement. For example, if the accuracy of a model decreases from 84% to 51%, the relative drift is calculated via (2):

$$D_{i,j} = \frac{0.51 - 0.84}{0.84} \times 100 \approx -39.29\% \quad (2)$$

Table 2 summarizes these parameters alongside the resulting drift rate computed using (2).

Table 2. Values used in the illustrative performance-drift calculation.

Quantity	Value used in the example
Earlier accuracy, P_i	84% (0.84)
Later accuracy, P_j	51% (0.51)
Relative performance drift, $D_{i,j}$	-39.29%

Because LLM performance is multidimensional, a single task-level drift measure may not adequately represent the overall behavioral change of a model [6]. Let $P_{k,i}$ and $P_{k,j}$ represent the performance of model versions i and j on evaluation dimension k , respectively. A weighted multi-dimensional drift score can then be expressed as (3):

$$D_{\text{total}} = \sum_{k=1}^n w_k \left(\frac{P_{k,j} - P_{k,i}}{P_{k,i}} \right) \quad (3)$$

where w_k is the importance weight assigned to dimension k , subject to the constraint in (4):

$$\sum_{k=1}^n w_k = 1, \quad w_k \geq 0. \quad (4)$$

The dimensions k in (3) may represent reasoning, coding, instruction following, truthfulness, safety, formatting, or multi-turn conversational performance [6]. This formulation reflects the multidimensional nature of LLM evaluation and allows organizations to assign greater importance to application-critical behaviors.

However, an observed difference between two model versions does not necessarily imply statistically significant behavioral drift [2]. Because LLM outputs may vary across repeated trials, the difference between versions should be evaluated using repeated measurements. Let $\bar{X}_i$ and $\bar{X}_j$ denote the mean performance scores of two model versions, with standard deviations s_i and s_j and sample sizes n_i and n_j . The observed performance difference is defined in (5):

$$\Delta P = \bar{X}_j - \bar{X}_i \quad (5)$$

A standardized statistic for comparing the two means can be expressed as (6):

$$Z = \frac{\bar{X}_j - \bar{X}_i}{\sqrt{\frac{s_i^2}{n_i} + \frac{s_j^2}{n_j}}} \quad (6)$$

where a large absolute value of Z in (6) provides evidence that the observed difference is larger than would typically be expected from sampling variation alone, subject to the assumptions of the statistical procedure. Therefore, repeated trials and statistical testing can improve the reliability of longitudinal drift detection.

4 CURRENT APPROACHES AND COMPARATIVE ANALYSIS

To effectively monitor and mitigate drift, researchers and engineers utilize several parallel approaches. Figure 3 outlines a continuous evaluation pipeline integrating behavioral fingerprinting and statistical testing based on the equations derived in Section 3.2.

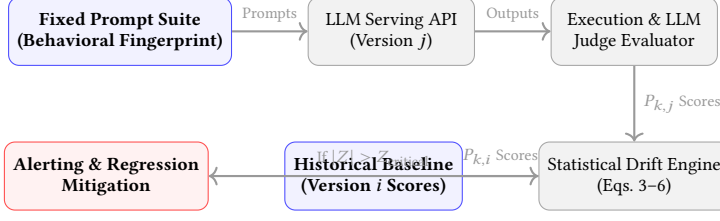


Fig. 3. Architecture of a Continuous Behavioral Fingerprinting and Drift Monitoring Pipeline.

4.1 Behavioral Fingerprinting

Behavioral fingerprinting involves maintaining a fixed collection of prompts that represent important model behaviors [2]. Each model version is evaluated using the same fingerprint, and its outputs are compared with historical results. A fingerprint may contain mathematical questions (Section 3.1), coding problems (Section 4.3), structured-output tasks, instruction-following tests, safety prompts, and multi-turn conversations [4].

4.2 Multi-Metric Evaluation

A major limitation of single-score benchmarking is that it can conceal trade-offs [6]. Liang et al. [6] demonstrate the usefulness of evaluating multiple dimensions simultaneously. A production drift-monitoring system can maintain separate metrics, as itemized in Table 3.

Table 3. Evaluation dimensions and example measurements.

Dimension	Example Measurement
Reasoning	Task accuracy
Coding	Unit-test pass rate
Instruction following	Constraint satisfaction
Truthfulness	TruthfulQA-style accuracy
Safety	Unsafe-output/refusal metrics
Formatting	Schema-valid output rate
Conversation	Multi-turn task completion
Human preference	Human or calibrated LLM judgment

This structure makes it possible to identify the specific dimensions affected by an update. Figure 4 illustrates the multidimensional comparison across dimensions k .

4.3 Executable Evaluation

Executable evaluation is particularly valuable for code-generation tasks [3]. Instead of comparing generated code to a reference solution, the generated program is executed against test cases [3]. HumanEval demonstrates the effectiveness of this approach [3]. Executable tests can therefore serve as automated regression tests for LLM applications.

5 RESEARCH CHALLENGES AND LIMITATIONS

5.1 Proprietary Model Opacity

The most significant challenge is the black-box nature of commercial LLMs [2]. Providers may not publicly disclose all changes to training data, model parameters, post-training procedures, safety

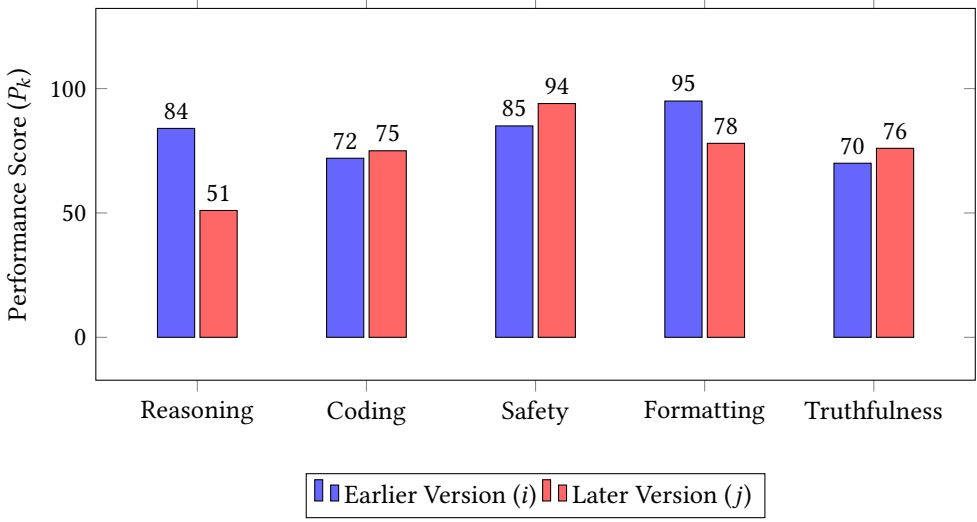


Fig. 4. Multidimensional performance drift across evaluation dimensions (represented by k in (3)) [2, 6]. A single model update can simultaneously result in improvements (e.g., Safety, Coding) and regressions (e.g., Reasoning, Formatting), highlighting the inadequacy of relying on a single static capability score.

systems, or routing mechanisms. Consequently, researchers can often identify that a behavioral change occurred without determining its exact cause.

5.2 Stochasticity and Benchmark Contamination

LLMs are generative systems, and outputs may differ across repeated trials [1]. This makes simple before-and-after comparisons unreliable without statistical metrics. Widely used benchmarks can also eventually become part of model-development pipelines or training data [5]. A combination of historical regression tests and fresh evaluation data provides stronger evidence about general behavioral change [6].

5.3 Benchmark Contamination

Widely used public benchmarks risk data leakage into pre-training corpora and alignment pipelines over time [6]. While fixed evaluation suites remain essential for deterministic regression testing, they should be continuously augmented with newly constructed, unseen task instances. Integrating static historical regression tests with dynamic evaluation datasets provides stronger empirical proof of genuine behavioral change rather than simple dataset memorization.

5.4 Lack of Ground Truth and Evaluation Ambiguity

Quantifying drift is straightforward for domains with deterministic ground truth, such as code compilation, unit testing, and mathematical execution. Conversely, subjective properties—including helpfulness, tone, contextual appropriateness, and safety—lack explicit ground truth. Human evaluation provides qualitative grounding but introduces high financial overhead and annotator variance. While automated LLM-as-a-judge approaches offer scalable evaluation, they introduce distinct measurement biases, including verbosity and positional bias [13].

5.5 Limitations of Aggregate Metrics

Macro-level aggregate metrics often obscure critical domain-specific regressions. A marginal decrease in a composite benchmark score can mask severe performance loss in a specialized task or output format crucial to a production application [2]. To ensure system stability, drift monitoring frameworks must track task-level and scenario-level metrics alongside global benchmark averages.

6 RESEARCH GAPS AND FUTURE DIRECTIONS

Despite recent advancements in quantifying model behavior over time, detecting, characterizing, and mitigating performance drift in deployed Large Language Models (LLMs) remains an open challenge. Current frameworks predominantly assess models in isolated environments, leaving significant gaps in understanding real-world behavioral dynamics. This section outlines six critical research frontiers necessary to establish robust, drift-resilient AI software architectures.

6.1 Longitudinal Multi-Turn Evaluation

Most conventional benchmarks evaluate LLM capabilities through single-turn prompt-response pairs [5, 6]. However, production deployments frequently rely on multi-turn dialogue, where context retention, state maintenance, and goal preservation are critical. Recent findings indicate that models exhibit cumulative drift over continuous interactions, leading to context degradation and goal erosion [4]. Future research must establish standardized, long-horizon evaluation protocols that measure conversational stability and state equilibrium across extended interaction sessions [13].

6.2 Standardized Drift Taxonomy

While empirical studies confirm that commercial APIs experience significant behavioral shifts over time [2], the research community lacks a unified nomenclature to categorize these changes. We propose structuring future empirical investigations around eight distinct operational dimensions of performance drift:

- (1) **Capability Drift:** Shifts in domain-specific problem-solving accuracy and core subject knowledge [5].
- (2) **Reasoning Drift:** Fluctuations in multi-step logical deduction, symbolic execution, and mathematical reasoning [2].
- (3) **Semantic Drift:** Alterations in output verbosity, stylistic tone, or lexical diversity.
- (4) **Formatting Drift:** Regressions in strict schema compliance, tag generation, or structured output parsing (e.g., JSON/XML) [12].
- (5) **Instruction-Following Drift:** Decreased adherence to system prompt constraints, operational guidelines, or negative constraints [10].
- (6) **Safety Drift:** Unintended shifts in alignment guardrails, resulting in either over-refusal of benign requests or altered safety thresholds [9].
- (7) **Truthfulness Drift:** Variations in factual grounding, hallucination propensity, or susceptibility to common misconceptions [7, 11].
- (8) **Multi-Turn Interaction Drift:** Memory decay, context window instability, or session goal drift across consecutive conversational turns [4].

Adopting a formal taxonomy will enable cross-study comparability and facilitate fine-grained monitoring of specific failure modes.

6.3 Predictive Evaluation and Pre-Deployment Probing

Current industrial monitoring is predominantly reactive: downstream developers detect model regressions only after an unannounced API update alters production behavior [2]. Future research

should focus on proactive evaluation pipelines that test candidate model updates against historical behavioral fingerprints prior to full deployment. Integrating continuous statistical testing framework—such as calculating significance test statistics (Equation ??)—allows automated staging environments to flag regression risks before model endpoints serve live traffic.

6.4 Time-Aware Benchmarking Infrastructure

A persistent confounding factor in longitudinal evaluation is benchmark instability. When evaluation codebases, prompt templates, or underlying test datasets undergo unannounced modifications, isolating genuine model behavior shifts from benchmark noise becomes difficult [6]. Future benchmarking suites must enforce immutable versioning, explicit time-stamping, frozen decoding parameters, and deterministic evaluation scripts to ensure reproducible longitudinal measurements.

6.5 Application-Level Regression Testing

General academic benchmarks (e.g., [3, 5]) provide broad signal but fail to capture application-specific execution constraints. Enterprise software systems require specialized regression testing suites derived directly from operational production workflows. For instance, code-generation pipelines require automated evaluation of contract compliance, schema validity, and integration test pass rates alongside traditional code completion metrics [12].

6.6 Drift in Agentic and Tool-Augmented Systems

As LLMs transition from standalone generation tools to autonomous agents that interface with external APIs, databases, and execution sandboxes, performance drift becomes non-linear. In tool-augmented architectures, a subtle shift in model reasoning or function-calling format can trigger cascading system failures even when underlying base language capabilities remain stable. Future research must evaluate how performance drift propagates through compound AI systems, tool-selection routines, and multi-agent coordination pipelines.

7 CONCLUSION

Performance drift represents an important reliability challenge for continuously updated Large Language Models [2]. Research demonstrates that LLM behavior can change across mathematical reasoning, programming, instruction following, truthfulness, safety, and multi-turn interaction [4]. Continuous, multidimensional evaluation frameworks—such as the fingerprinting pipeline shown in Figure 3 alongside statistical testing in (3) and (6)—are essential to ensure that downstream enterprise applications remain reliable as underlying models evolve [2].

REFERENCES

- [1] Emily M. Bender, Timnit Gebru, Alisha McMillan-Major, and Shmargaret Shmitchell. 2021. On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FACt)*. 610–623. doi:10.1145/3442188.3445922
- [2] Lingjiao Chen, Matei Zaharia, and James Zou. 2024. How Is ChatGPT’s Behavior Changing Over Time? *Harvard Data Science Review* 6, 2 (2024). doi:10.1162/99608f92.5317da47
- [3] Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, Alex Ray, Raul Puri, Gretchen Krueger, Michael Petrov, Heidy Khlaaf, Girish Sastry, Pamela Mishkin, Brooke Chan, Scott Gray, Nick Ryder, Mikhail Pavlov, Alethea Power, Lukasz Kaiser, Mohammad Bavarian, Clemens Winter, Philippe Tillet, Felipe Petroski Such, Dave Cummings, Matthias Plappert, Fotios Chantzis, Elizabeth Barnes, Ariel Herbert-Voss, William Hebgen Guss, Alex Nichol, Alex Paino, Nikolas Tezak, Jie Tang, Igor Babuschkin, Suchir Balaji, Shantanu Jain, William Saunders, Christopher Hesse, Andrew N. Carr, Jan Leike, Joshua Achiam, Vedant Misra, Evan Morikawa, Alec Radford, Matthew Knight, Miles Brundage, Mira Mayer, Peter Welinder, Bob McGrew, Dario Amodei, Wojciech Zaremba, and Ilya Sutskever. 2021. Evaluating Large Language Models Trained on Code. *arXiv preprint arXiv:2107.03374* (2021). doi:10.48550/arXiv.2107.03374

- [4] Vaibhav Dongre, Ryan A. Rossi, Viet Dac Lai, et al. 2025. Drift No More? Context Equilibria in Multi-Turn LLM Interactions. *arXiv preprint arXiv:2510.07777* (2025). doi:10.48550/arXiv.2510.07777
- [5] Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. 2021. Measuring Massive Multitask Language Understanding. In *International Conference on Learning Representations (ICLR)*.
- [6] Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narayanan, Yuhui Wu, Ananya Kumar, Benjamin Newman, Binhang Yuan, Benjamin Yan, Ce Zhang, Christian Cosgrove, Christopher D. Manning, Christopher Ré, Diana Acosta-Navas, Drew A. Hudson, Eric Zelikman, Esin Durmus, Daniel E. Ho, Emily Lee, Esin Stafford, Frank Teng, Freda M. Huang, et al. 2022. Holistic Evaluation of Language Models. *arXiv preprint arXiv:2211.09110* (2022). doi:10.48550/arXiv.2211.09110
- [7] Stephanie Lin, Jacob Hilton, and Owain Evans. 2022. TruthfulQA: Measuring How Models Mimic Human Falsehoods. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*. 3214–3252. doi:10.18653/v1/2022.acl-long.229
- [8] Meng Liu, Ruixi Liu, Yujia Zhu, et al. 2024. A Survey on the Real Power of ChatGPT. *arXiv preprint arXiv:2405.00704* (2024). doi:10.48550/arXiv.2405.00704
- [9] OpenAI, Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altschmidt, Sam Altman, Shyamal Anadkat, Igor Avila, Igor Babuschkin, Suchir Balaji, Valerie Balcom, Paul Baltescu, Haiming Bao, Mohammad Barnes, Nikolay Bashlykov, et al. 2023. GPT-4 Technical Report. *arXiv preprint arXiv:2303.08774* (2023). doi:10.48550/arXiv.2303.08774
- [10] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. 2022. Training Language Models to Follow Instructions with Human Feedback. *Advances in Neural Information Processing Systems (NeurIPS)* 35 (2022), 27730–27744. doi:10.52202/068431-2011
- [11] Kellogg Peirine, Anne Imouza, Charles Thibault, et al. 2023. Towards Reliable Misinformation Mitigation: Generalization, Uncertainty, and GPT-4. *arXiv preprint arXiv:2305.14928* (2023). doi:10.48550/arXiv.2305.14928
- [12] Haoye Tian, Weiqi Lu, Tsz-On Li, et al. 2023. Is ChatGPT the Ultimate Programming Assistant – How Far Is It? *arXiv preprint arXiv:2304.11938* (2023). doi:10.48550/arXiv.2304.11938
- [13] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*. doi:10.48550/arXiv.2306.05685